# Wireless 02 | Guest Wireless with AGNI

## Overview

In this lab, you'll configure a guest wireless experience using AGNI for captive portal authentication and CV-CUE for SSID management. We'll also set up VXLAN tunneling to isolate guest traffic.

**What you'll learn:**

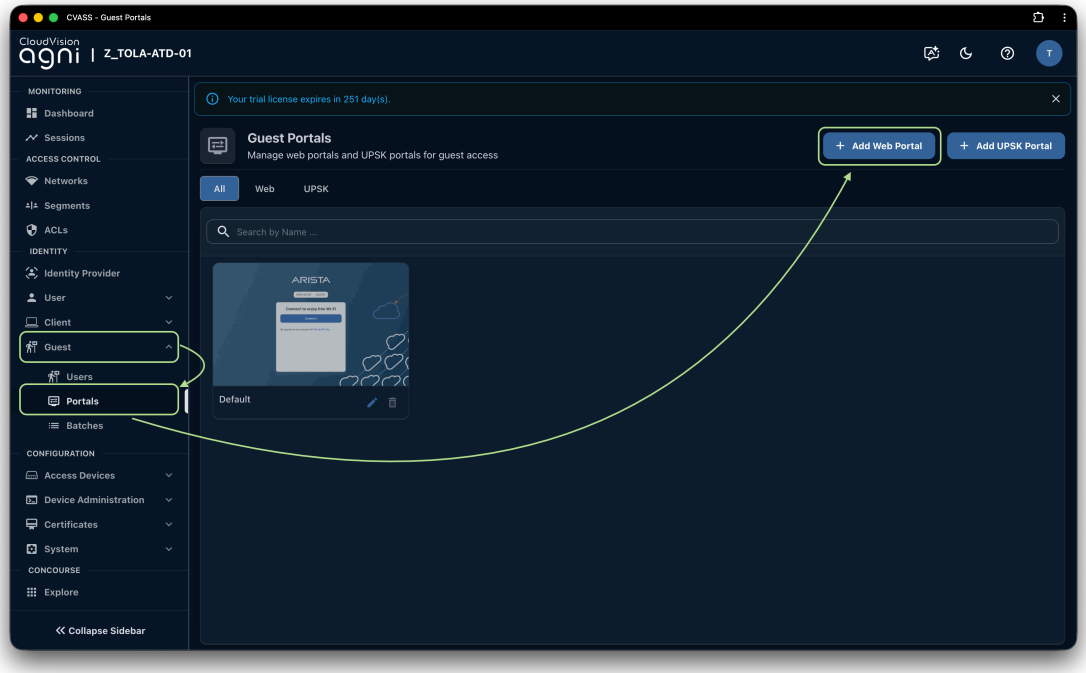
- Create a guest captive portal in AGNI
- Configure role profiles and segments for guest access
- Set up a guest SSID with RADIUS MAC authentication
- Tunnel guest traffic using VXLAN L2 tunnels

**Open Launchpad** | **Login Help** | **Lab Topology**

## AGNI Guest Captive Portal

Let's configure a Guest Captive Portal using AGNI for wireless clients. To configure the guest portal, you must configure both AGNI and CV-CUE.

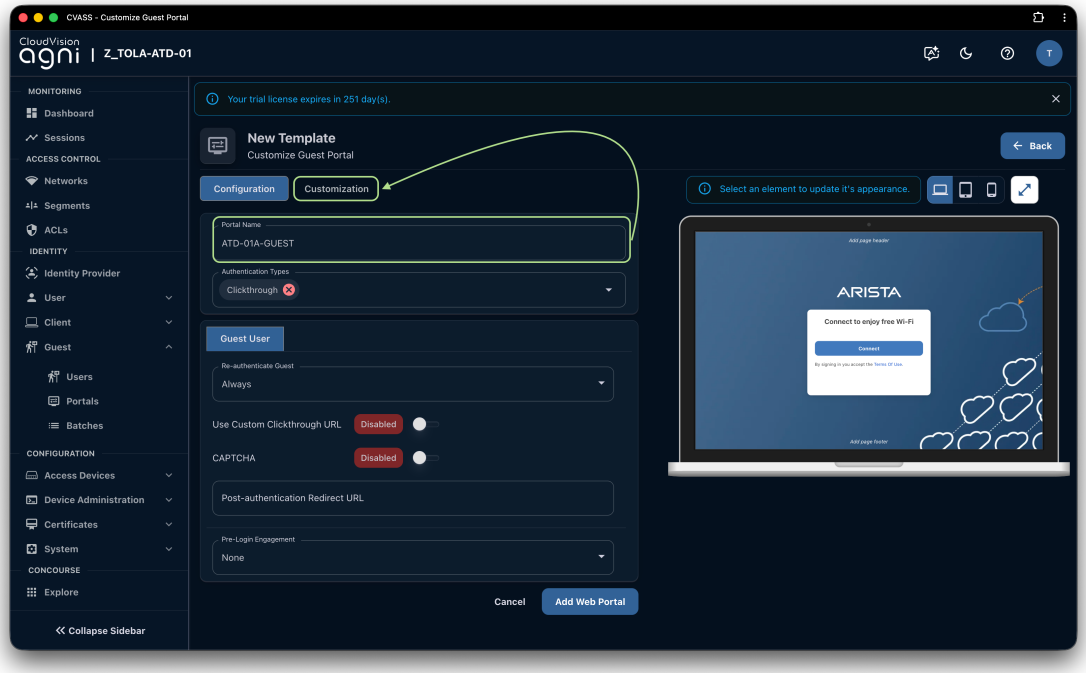
1. Navigate to `Guest > Portals` under the section `Identity`.




2. Select `+ Add Guest Portal` and configure the following

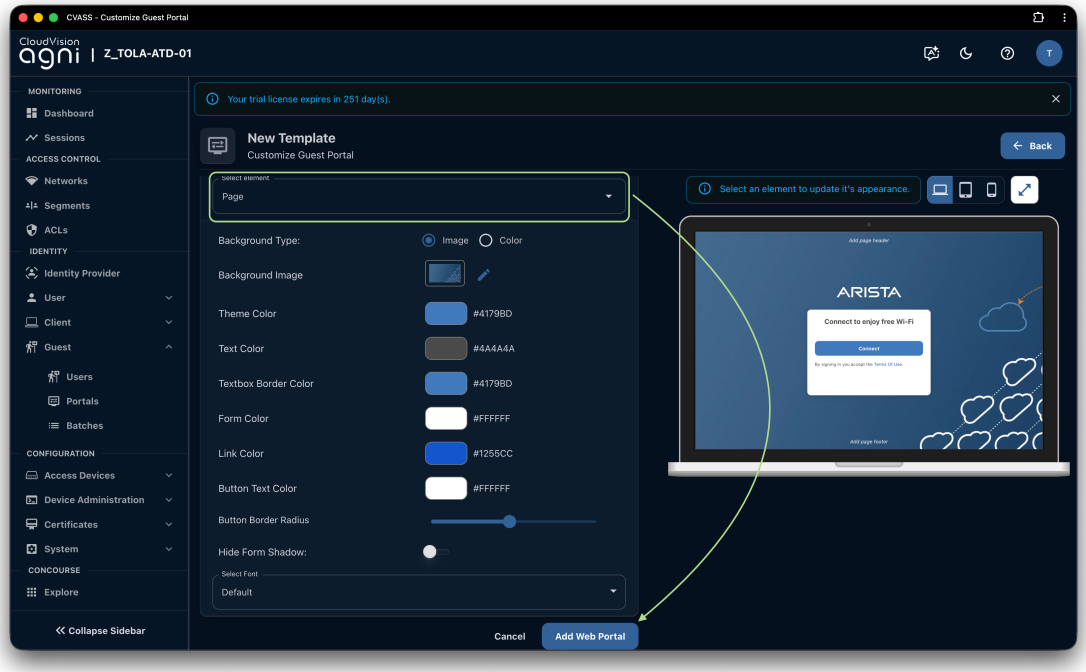
### Network Settings

| Field | Student 1 | Student 2 |
|---|---|---|
| Portal Name | `ATD-##A-GUEST` | `ATD-##B-GUEST` |
| Authentication Types | Clickthrough | Clickthrough |
| Re-Authenticate Guest | Always | Always |
| CAPTCHA | Disabled | Disabled |



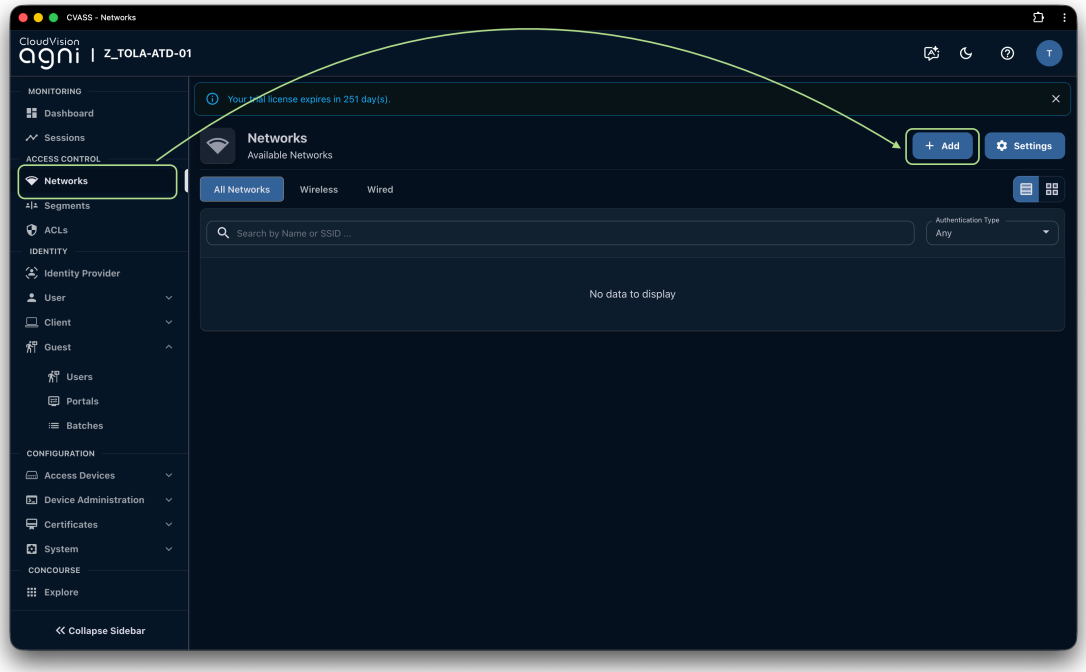

3. Select the `Customization` tab to customize the portal settings, and notice the elements. When done, select `Add Guest Portal`. The portal gets listed in the portal listing.

   - Page
   - Login Toggle
   - Terms of Use and Privacy Policy
   - Logo
   - Guest Login Submit Button
   - Etc




4. Select `Back`

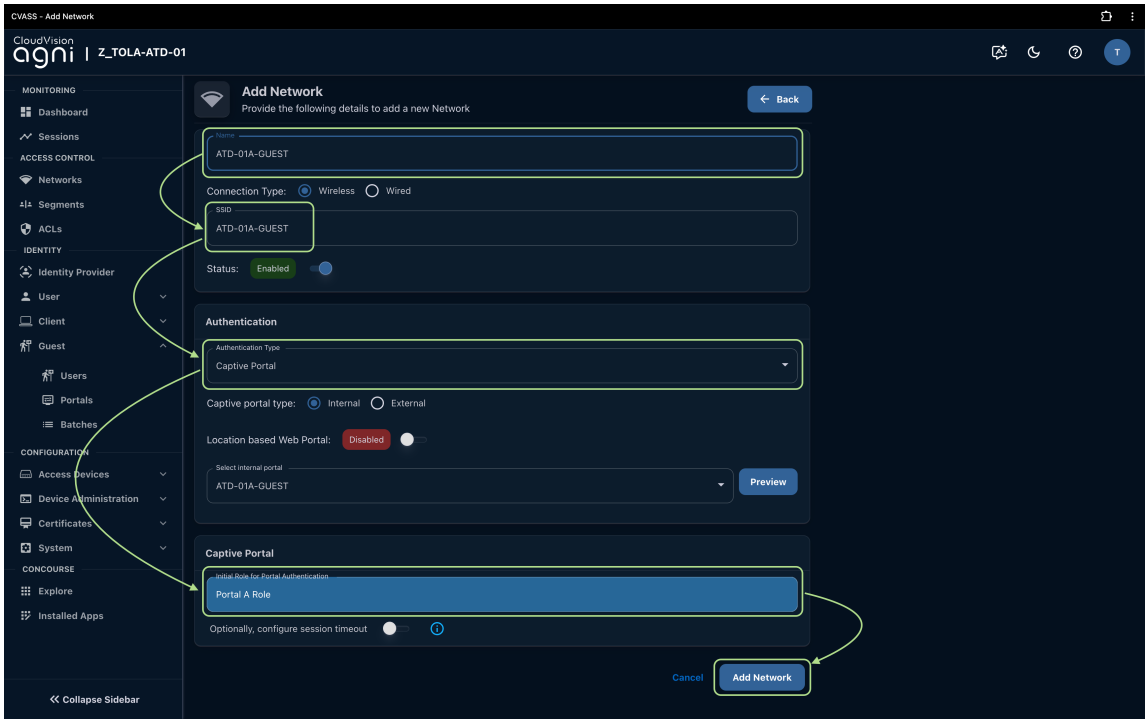
5. Navigate to the `Networks` under the section `Access Control`. Select `+ Add`




6. Add a new network with following settings

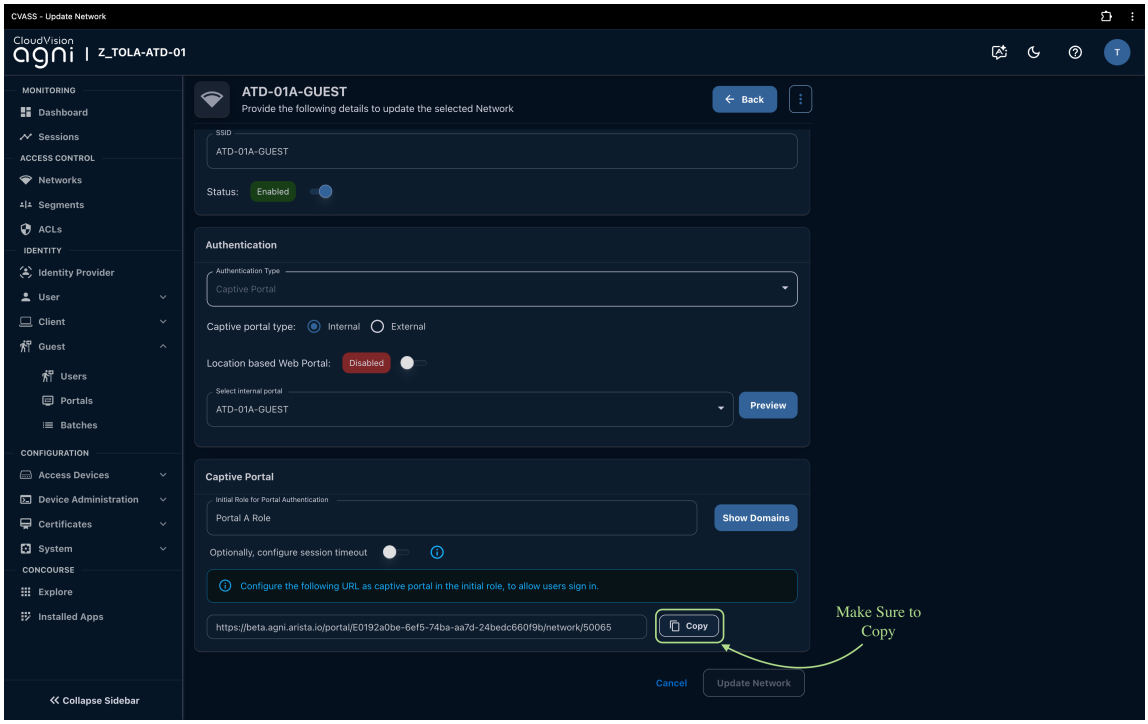
## Network Settings

| Field | Student 1 | Student 2 |
|---|---|---|
| Name | `ATD-##A-GUEST` | `ATD-##B-GUEST` |
| Connection Type | Wireless | Wireless |
| SSID | `ATD-##A-GUEST` | `ATD-##B-GUEST` |
| Authentication Type | Captive Portal | Captive Portal |
| Captive Portal Type | Internal | Internal |
| Select Internal Portal | `ATD-##A-GUEST` | `ATD-##B-GUEST` |
| Internal Role for Portal Authentication | `Portal A Role` | `Portal B Role` |




7. Select `Add Network`.

8. Copy the portal URL at the bottom of the page.




**AGNI URL**

Make sure to **copy the AGNI Guest Portal URL**, we are going to use this in CV-CUE for Guest Portal Redirection.

Log in to **CV-CUE** from the Launchpad if you're not already there.

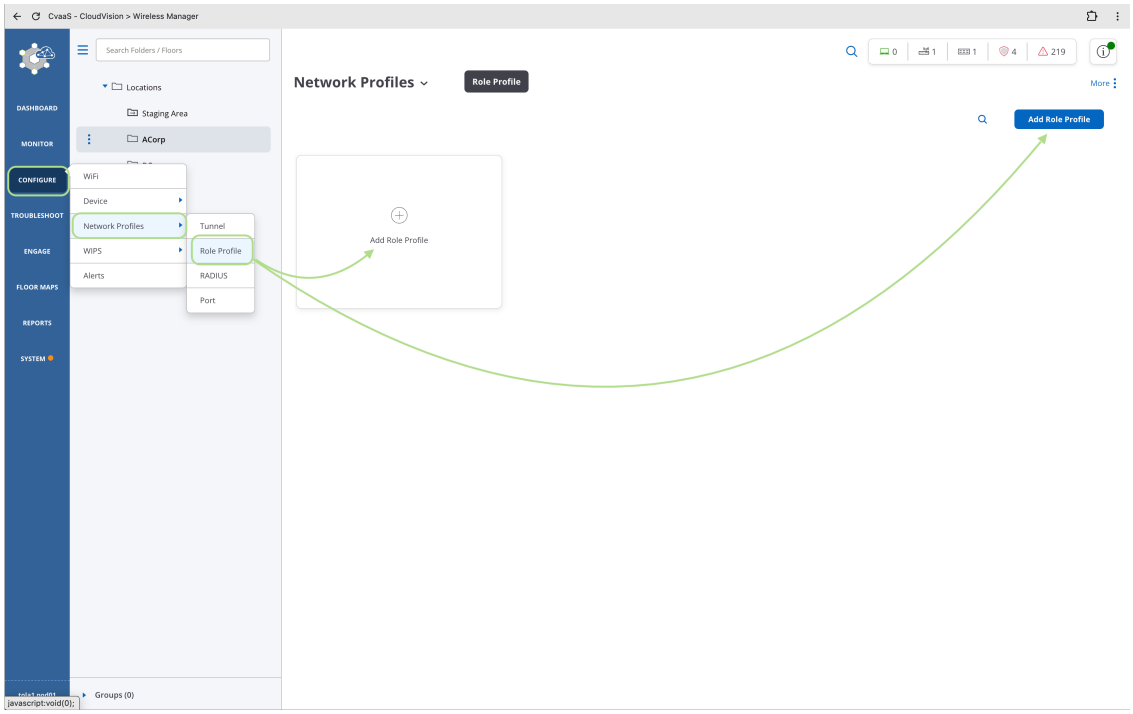
## Role Profile (CV-CUE)

Let's configure two role profiles and the SSID settings. This will ensure the guest SSID we create is enabled with redirection to the AGNI captive portal. These roles are:

- Portal (A/B) Role: This role will be assigned initially to ensure captive portal redirection
- Guest (A/B) Role: This role will be assigned by AGNI to drop our client into a "Guest Role" where we can further define policy

### Portal Role Profile

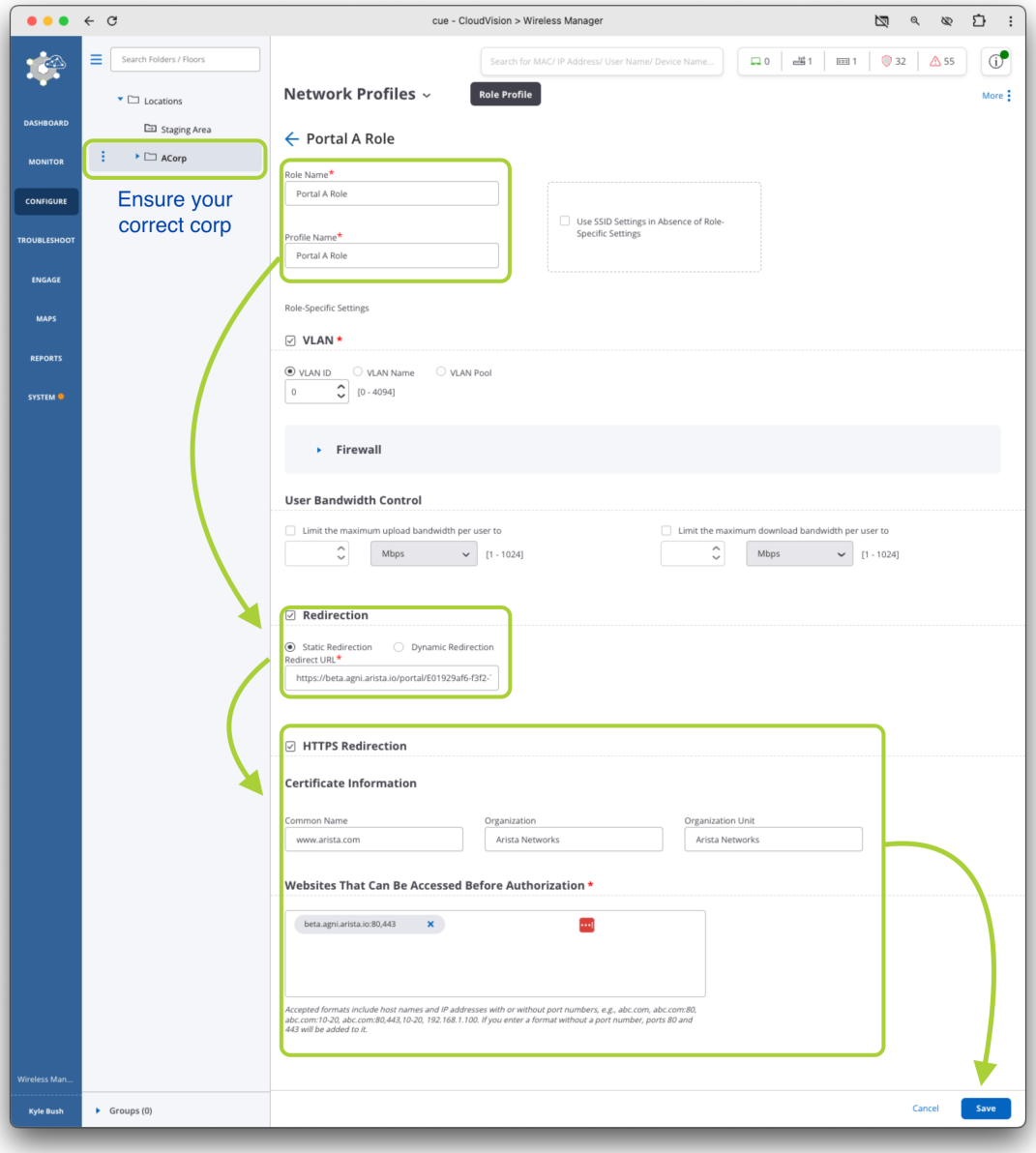
1. Log in to CV-CUE and navigate to `Configure > Network Profiles > Role Profile`.




2. Add a `Role Profile` using the setting below

**Network Settings**

| Field | Student 1 | Student 2 |
|---|---|---|
| Role Name | `Portal A Role` | `Portal B Role` |
| Profile Name | `Portal A Role` | `Portal B Role` |
| Redirection | Enabled | Enabled |
| Redirection Type | Static Redirection | Static Redirection |
| Redirect URL | `<Copied from AGNI>` | `<Copied from AGNI>` |
| HTTPS Redirection | Enabled | Enabled |
| Common Name | www.arista.com | www.arista.com |
| Organization | Arista Networks | Arista Networks |
| Organization Unit | Arista Networks | Arista Networks |




3. Select `Save` at the bottom of the page.

## Guest Role Profile

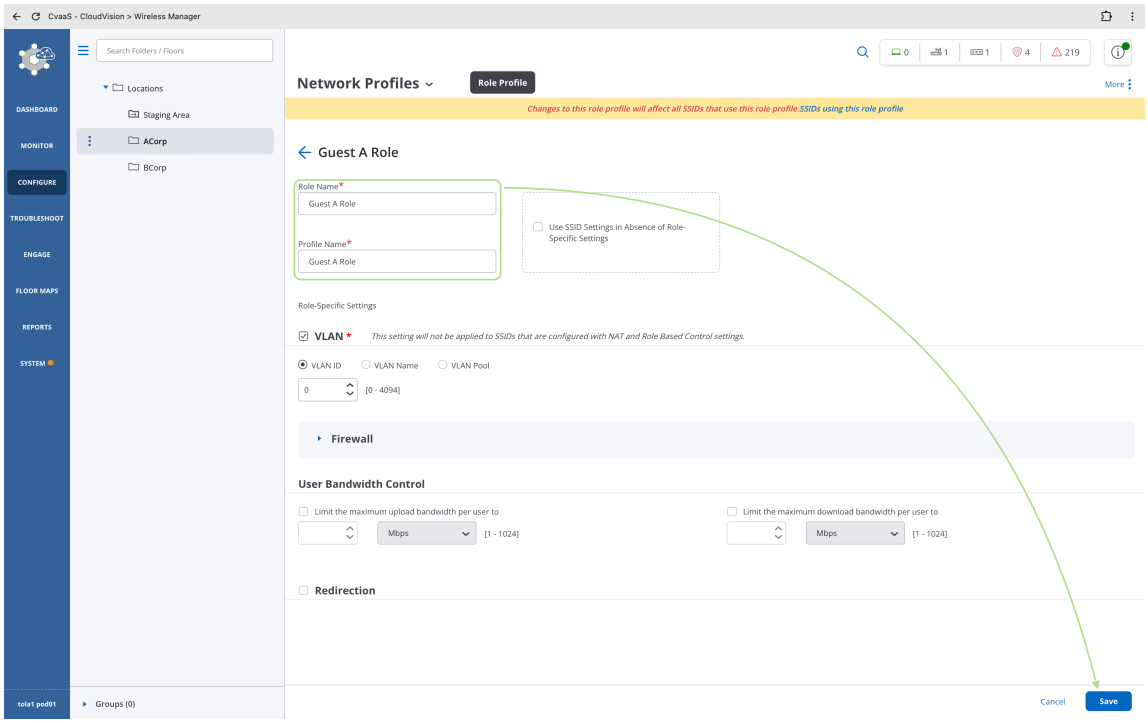
Next, we'll configure a Guest Role in CV-CUE to assign to Guest Users post authentication.

1. In CV-CUE, navigate back to `Configure > Network Profiles > Role Profile`.
2. Select `Add Role Profile`.
3. Add the `Role Name` as follows. This role is simple, but see the additional information below to

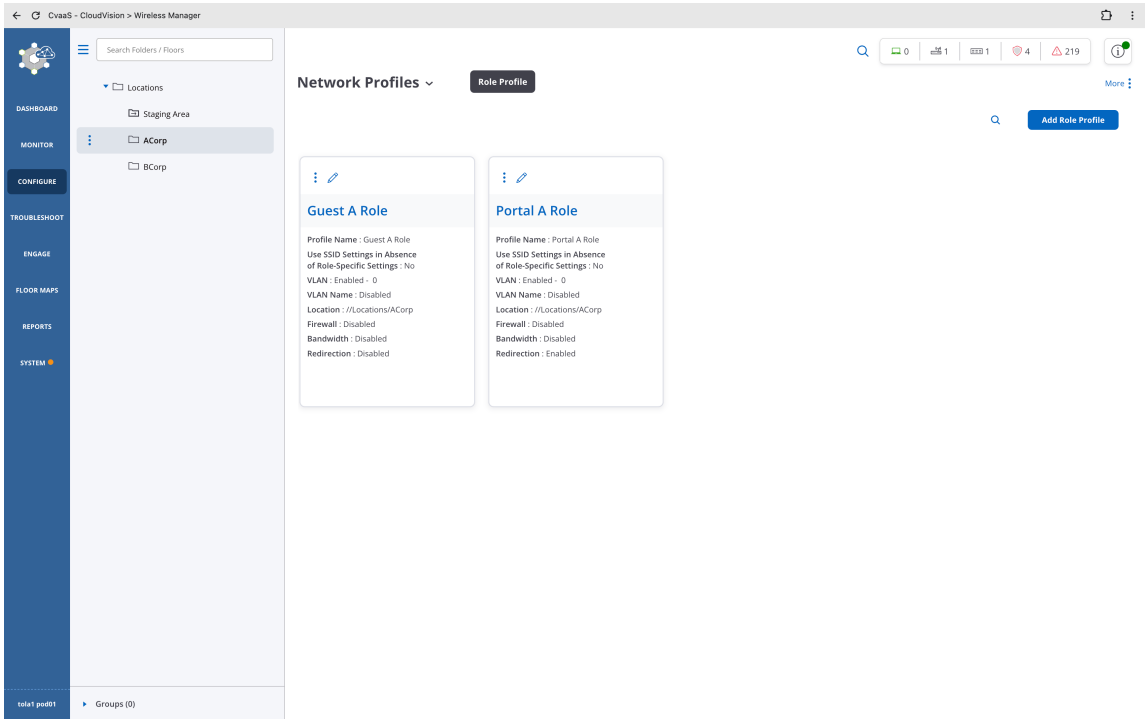
explore some of the options you have with roles.

**Network Settings**

| Field | Student 1 | Student 2 |
| --- | --- | --- |
| Role Name | `Guest A Role` | `Guest B Role` |




4. Select `Save` at the bottom of the page
5. You should now have two roles. We will refer back to these roles in both the AGNI and CUE SSID configuration.




## Additional Information

### VLAN

In this lab the VLAN is set to 0. In production networks you would define the Guest VLAN ID or Name that you want to assign to the Guest Users.

### Firewall

Layer 3-4 and Application Firewall Rules can be assigned to the Guest User Role.

### User Bandwidth Control

Upload and Download Bandwidth Limits can be assigned to the Guest User Role.

# Portal Segment (AGNI)

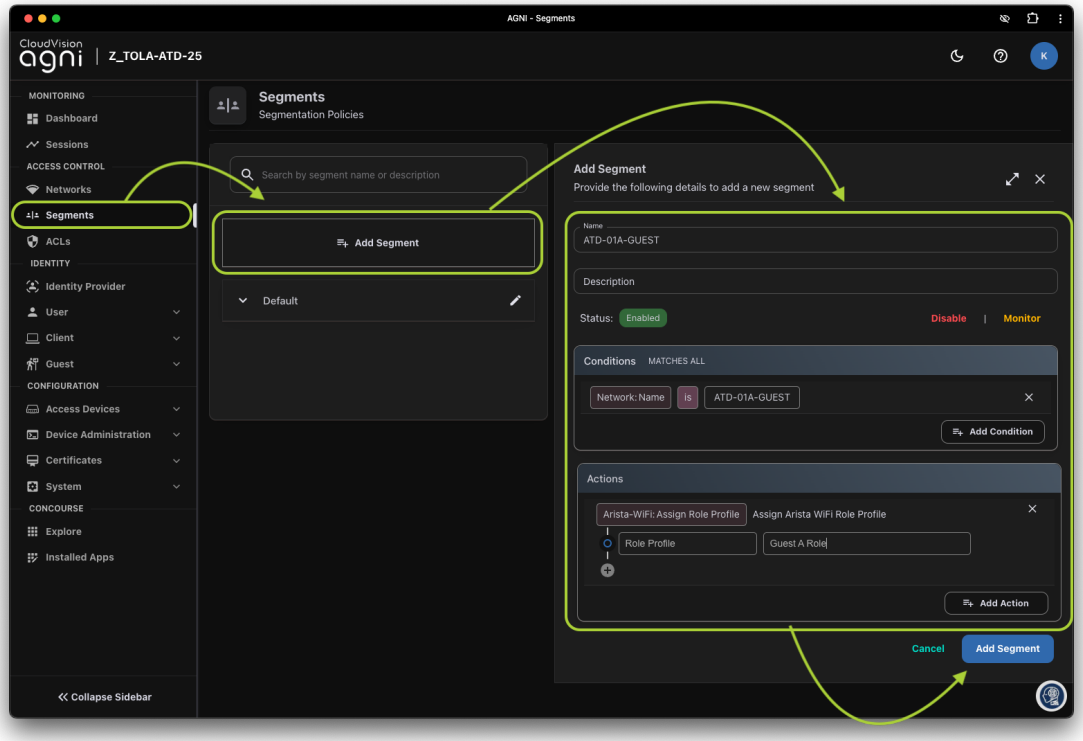
Next, we'll configure a Segment in AGNI to assign the Guest Role Profile post authentication.

1. Go back to AGNI and navigate to the `Access Control > Segments`.
2. Add a new Segment by selecting `+ Add Segment`

**Conditions for segments must MATCH ALL conditions line by line.**

**Network Settings**

| Field | Student 1 | Student 2 |
|---|---|---|
| Name | `ATD-##A-GUEST` | `ATD-##B-GUEST` |
| Condition | `Network:Name is ATD-##A-GUEST` | `Network:Name is ATD-##B-GUEST` |
| Action | `Arista-Wifi:Assign Role Profile` | `Arista-Wifi:Assign Role Profile` |
| Action Role Profile | `Guest A Role` | `Guest B Role` |




3. Select `Add Segment` at the bottom of the page.
4. Let's now add the Guest SSID

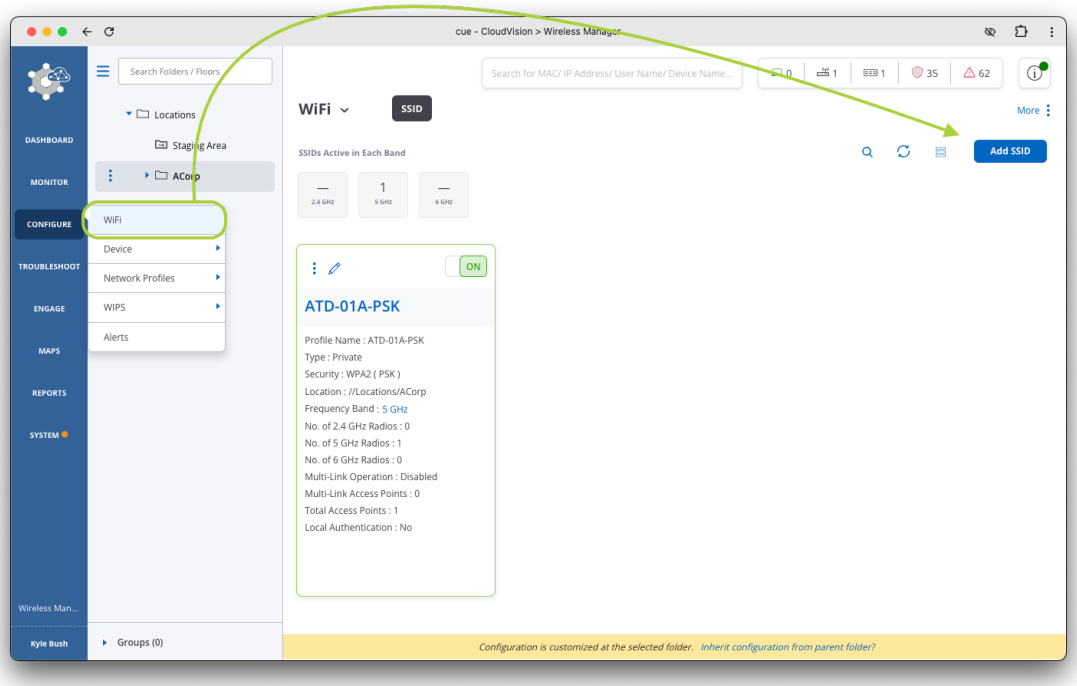
## Guest Portal SSID (CV-CUE)

Lastly, we'll configure and enable the Guest Captive Portal SSID and assign the pre and post authentication roles.

1. Let's navigate back to CV-CUE
2. Select the correct location `ACorp` (Student 1) or `BCorp` (Student 2)

> **CV-CUE Locations**
>
> Reminder, make sure you have selected your appropriate corporation before creating the SSID!

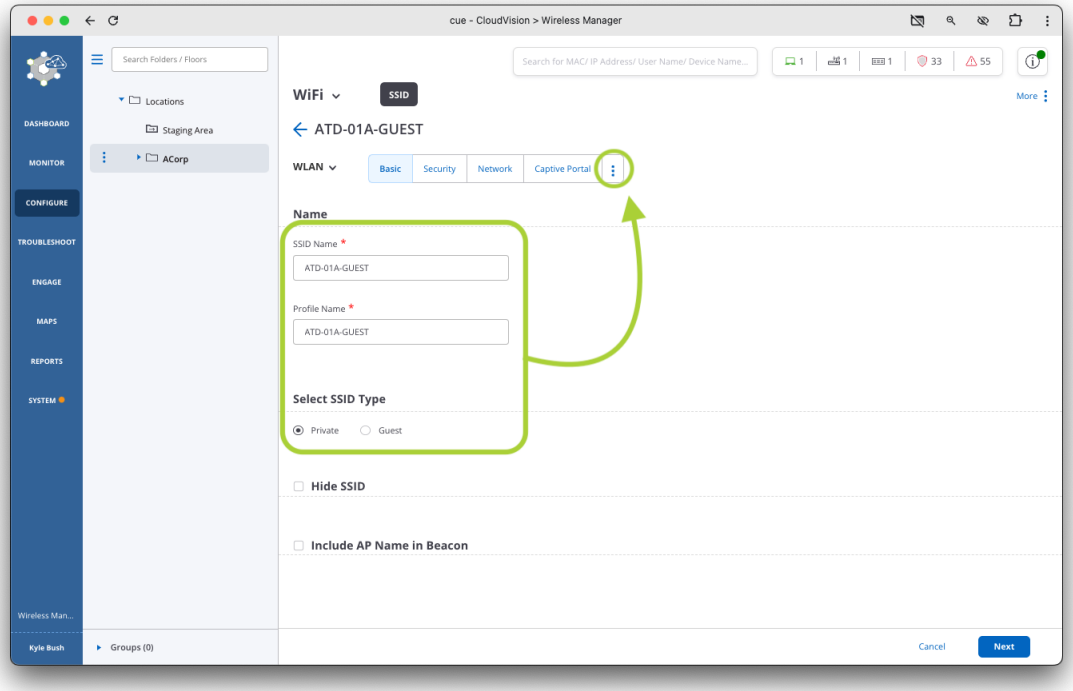
3. Navigate to `Configure > WiFi` and select `Add SSID`




4. Configure the SSID `Basic` settings using the settings below

**SSID Basic Settings**

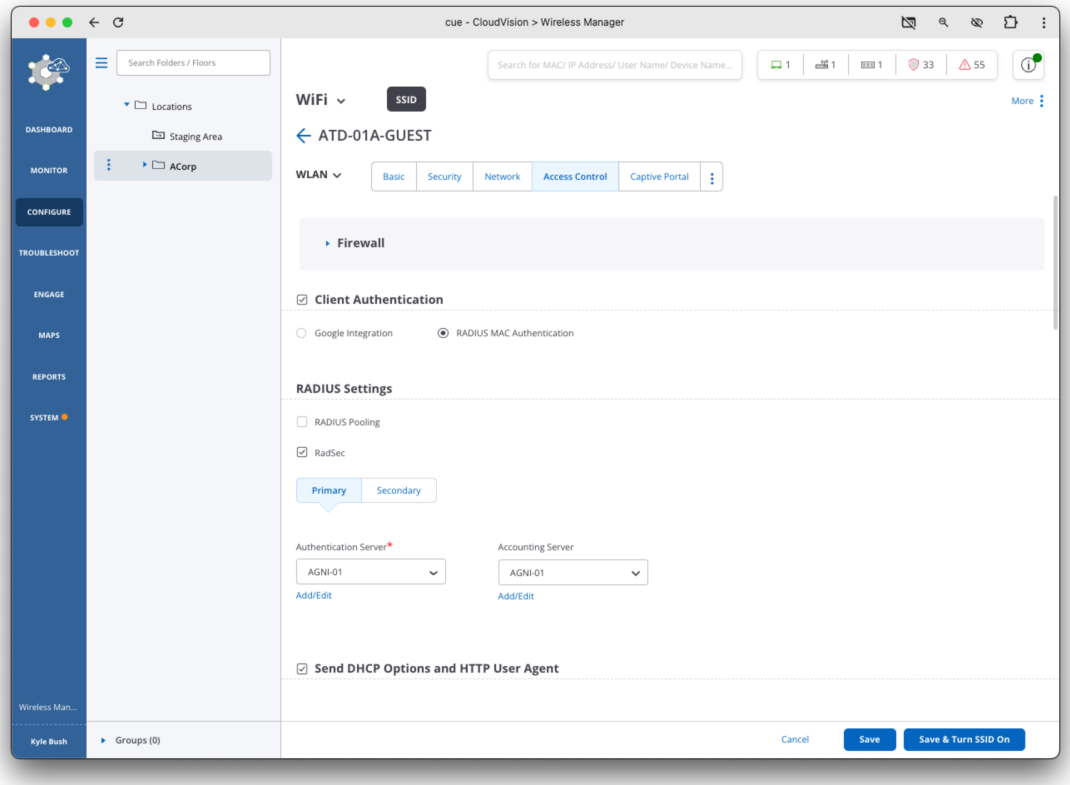
| Setting | Student 1 | Student 2 |
|---|---|---|
| SSID Name | `ATD-##A-GUEST` | `ATD-##B-GUEST` |
| SSID Type | Private | Private |

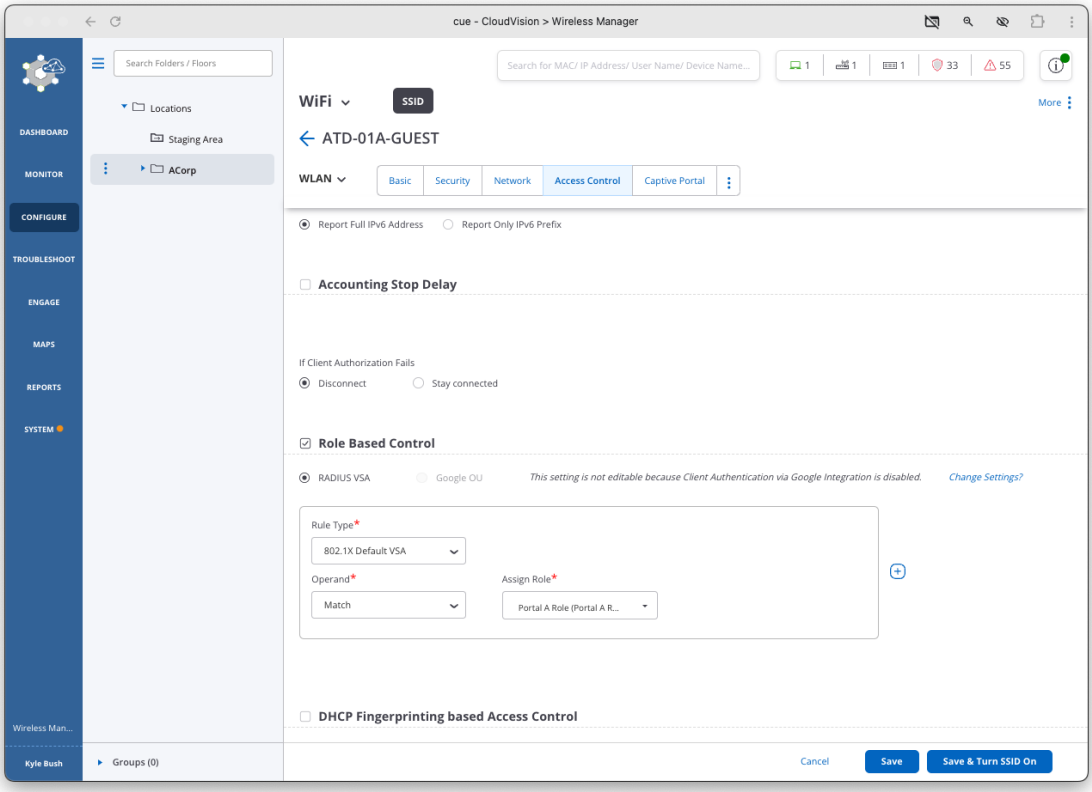



5. Select the 3 dots and select the `Access Control` tab and configure using the settings below

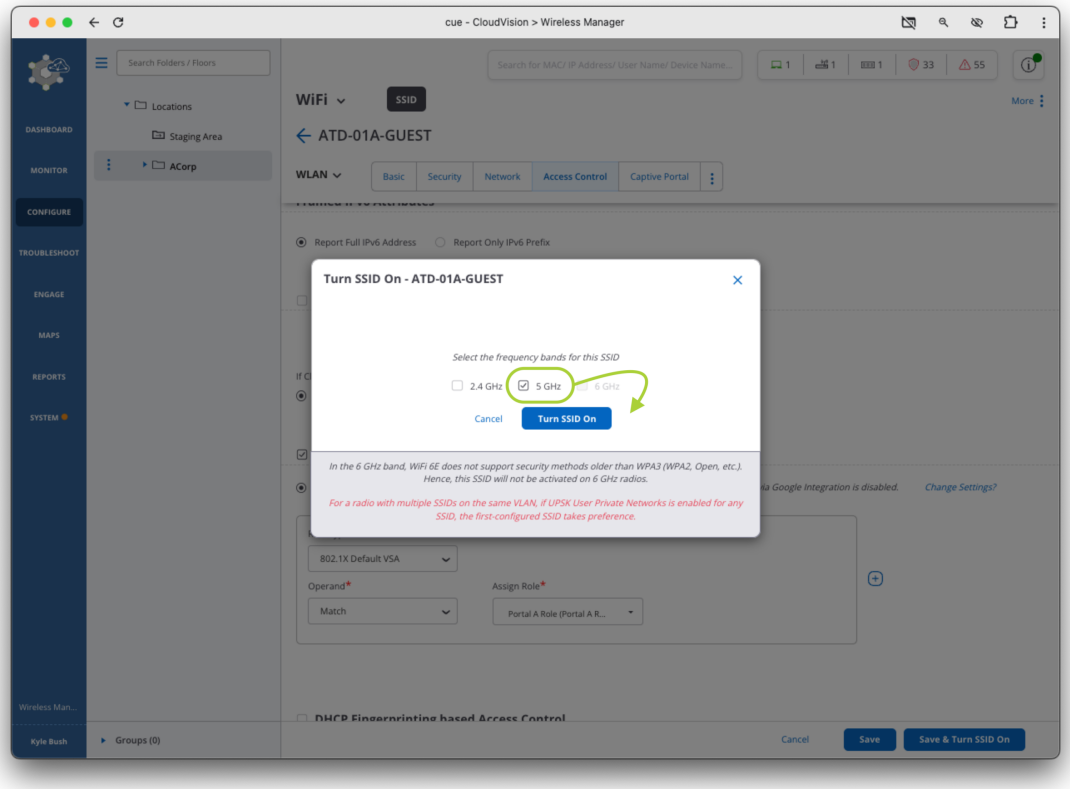
**SSID Access Control Settings**

| Setting | Value | Notes |
|---|---|---|
| **Client Authentication** | ☑ | |
| Authentication Type | Radius MAC Authentication | |
| **RadSec** | ☑ | |
| Authentication Server | `AGNI-01` | AGNI Radsec Server was configured already |
| Accounting Server | `AGNI-01` | |
| **Send DHCP Options and HTTP User Agent** | ☑ | |
| **Role Based Control** | ☑ | |
| Rule Type | 802.1X Default VSA | |
| Operand | Match | |
| Assign Role | Select All | Created in previous section |


cue - CloudVision > Wireless Manager
Search Folders / Floors
Locations
Staging Area
ACorp
DASHBOARD
MONITOR
CONFIGURE
TROUBLESHOOT
ENGAGE
MAPS
REPORTS
SYSTEM
Wireless Man...
Kyle Bush
Groups (0)
Search for MAC/ IP Address/ User Name/ Device Name...
1
1
1
33
55
WiFi
SSID
More
ATD-01A-GUEST
WLAN
Basic
Security
Network
Access Control
Captive Portal
Firewall
Client Authentication
Google Integration
RADIUS MAC Authentication
RADIUS Settings
RADIUS Pooling
RadSec
Primary
Secondary
Authentication Server*
AGNI-01
Add/Edit
Accounting Server
AGNI-01
Add/Edit
Send DHCP Options and HTTP User Agent
Cancel
Save
Save & Turn SSID On


cue - CloudVision > Wireless Manager
Search Folders / Floors
Locations
Staging Area
ACorp
DASHBOARD
MONITOR
CONFIGURE
TROUBLESHOOT
ENGAGE
MAPS
REPORTS
SYSTEM
Wireless Man...
Kyle Bush
Groups (0)
Search for MAC/ IP Address/ User Name/ Device Name...
1
1
1
33
55
WiFi
SSID
More
ATD-01A-GUEST
WLAN
Basic
Security
Network
Access Control
Captive Portal
Report Full IPv6 Address
Report Only IPv6 Prefix
Accounting Stop Delay
If Client Authorization Fails
Disconnect
Stay connected
Role Based Control
RADIUS VSA
Google OU
This setting is not editable because Client Authentication via Google Integration is disabled.
Change Settings?
Rule Type*
802.1X Default VSA
Operand*
Match
Assign Role*
Portal A Role (Portal A R...
DHCP Fingerprinting based Access Control
Cancel
Save
Save & Turn SSID On

6. Once all settings have been set, select the `Save & Turn SSID On` button at the bottom of the page.
7. You will be prompted to customize before enabling, select the `5 GHz` option on the next screen (un-check the `2.4 GHz` box if it's checked), then select `Turn SSID On`.

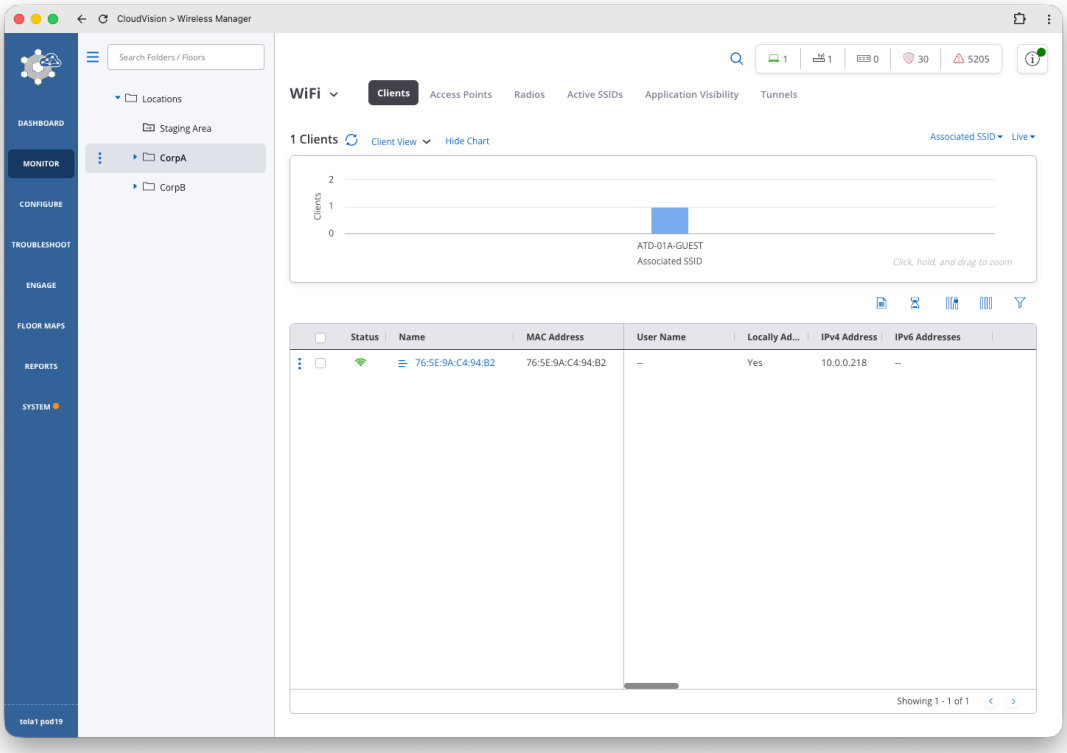


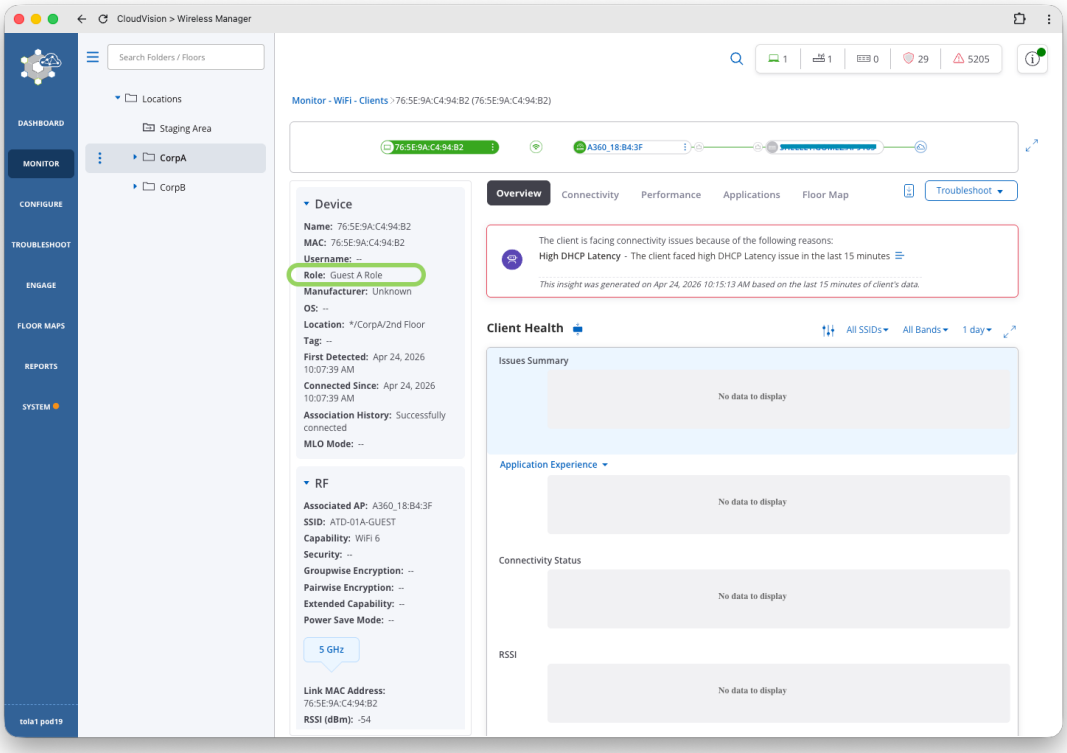

8. Join the new Wi-Fi and verify connectivity in CUE and AGNI

> **Join the guest Wi-Fi!**
>
> Give it a moment for the new SSID to come up, but once it's up try and join the Wi-Fi! You should be prompted with a captive portal.

**CV-CUE**

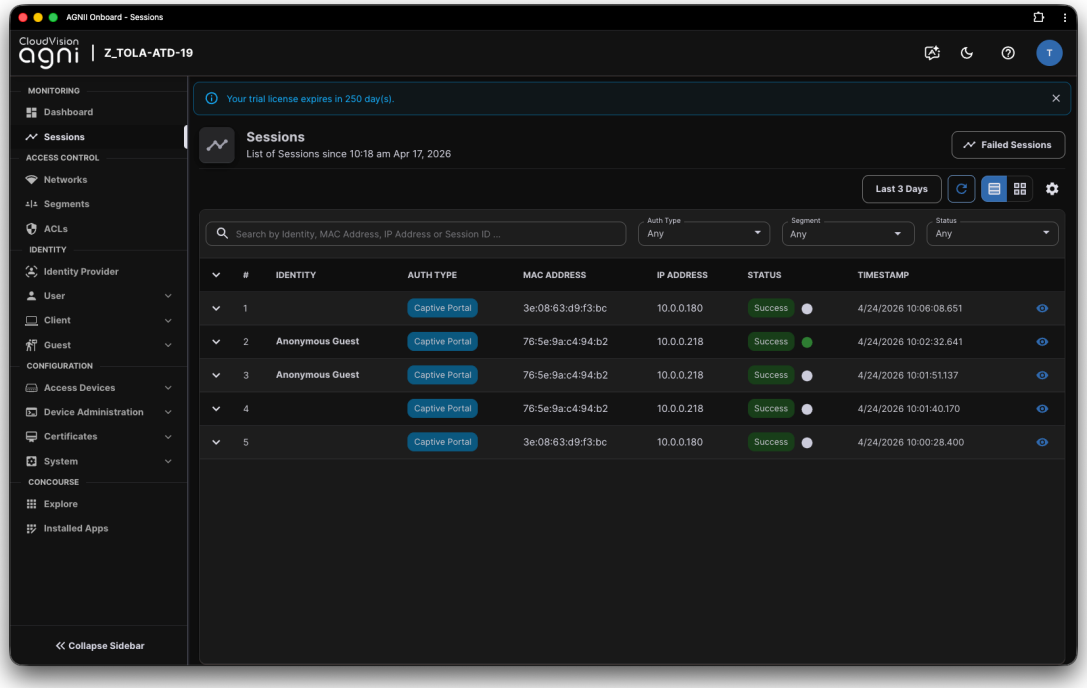
Navigate to `Monitor > Clients` in CV-CUE

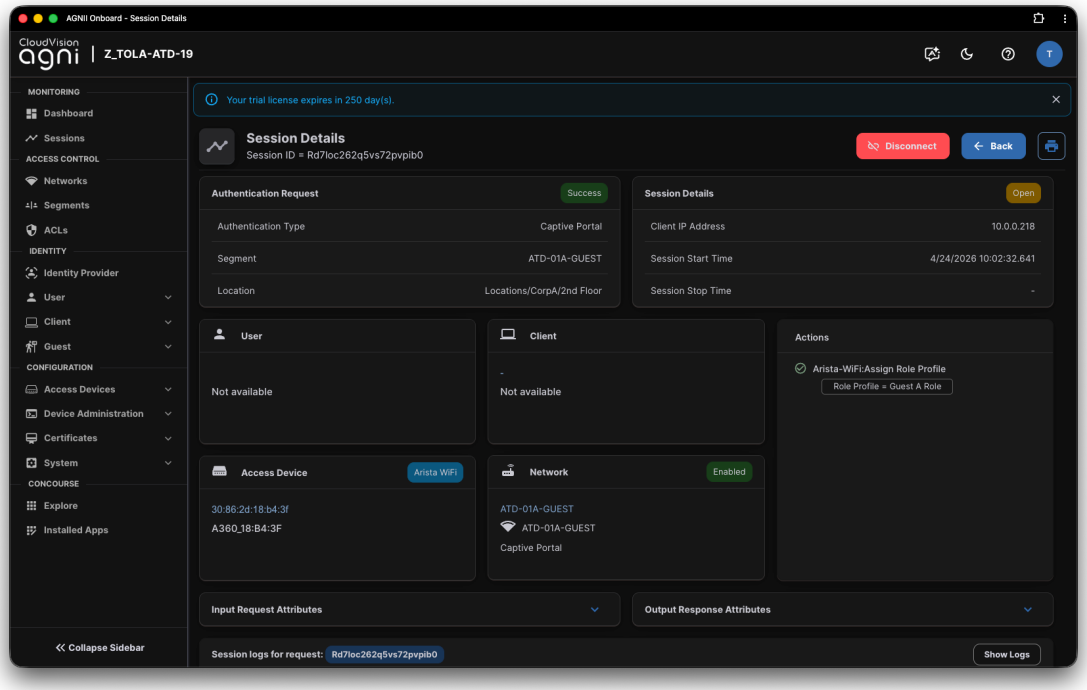

CloudVision > Wireless Manager
Search Folders / Floors
Locations
Staging Area
CorpA
CorpB
DASHBOARD
MONITOR
CONFIGURE
TROUBLESHOOT
ENGAGE
FLOOR MAPS
REPORTS
SYSTEM
tola1 pod19
WiFi
Clients
Access Points
Radios
Active SSIDs
Application Visibility
Tunnels
1 Clients
Client View
Hide Chart
Associated SSID
Live
Clients
ATD-01A-GUEST
Associated SSID
Click, hold, and drag to zoom
Status
Name
MAC Address
User Name
Locally Ad...
IPv4 Address
IPv6 Addresses
76:5E:9A:C4:94:B2
Yes
10.0.0.218
Showing 1 - 1 of 1


CloudVision > Wireless Manager
Monitor - WiFi - Clients > 76:5E:9A:C4:94:B2 (76:5E:9A:C4:94:B2)
A360_18:B4:3F
Device
Name: 76:5E:9A:C4:94:B2
MAC: 76:5E:9A:C4:94:B2
Username: --
Role: Guest A Role
Manufacturer: Unknown
OS: --
Location: */CorpA/2nd Floor
Tag: --
First Detected: Apr 24, 2026 10:07:39 AM
Connected Since: Apr 24, 2026 10:07:39 AM
Association History: Successfully connected
MLO Mode: --
RF
Associated AP: A360_18:B4:3F
SSID: ATD-01A-GUEST
Capability: WiFi 6
Security: --
Groupwise Encryption: --
Pairwise Encryption: --
Extended Capability: --
Power Save Mode: --
5 GHz
Link MAC Address: 76:5E:9A:C4:94:B2
RSSI (dBm): -54
Overview
Connectivity
Performance
Applications
Floor Map
Troubleshoot
The client is facing connectivity issues because of the following reasons:
High DHCP Latency - The client faced high DHCP Latency issue in the last 15 minutes
This insight was generated on Apr 24, 2026 10:15:13 AM based on the last 15 minutes of client's data.
Client Health
All SSIDs
All Bands
1 day
Issues Summary
No data to display
Application Experience
Connectivity Status
RSSI

## AGNI

Navigate to `Sessions` in AGNI



# Tunneling Guest Traffic

To highlight Arista's Controller-less Architecture, we will use the lab's Core 720DP to simulate a DMZ switch configured with VXLAN to tunnel guest traffic.

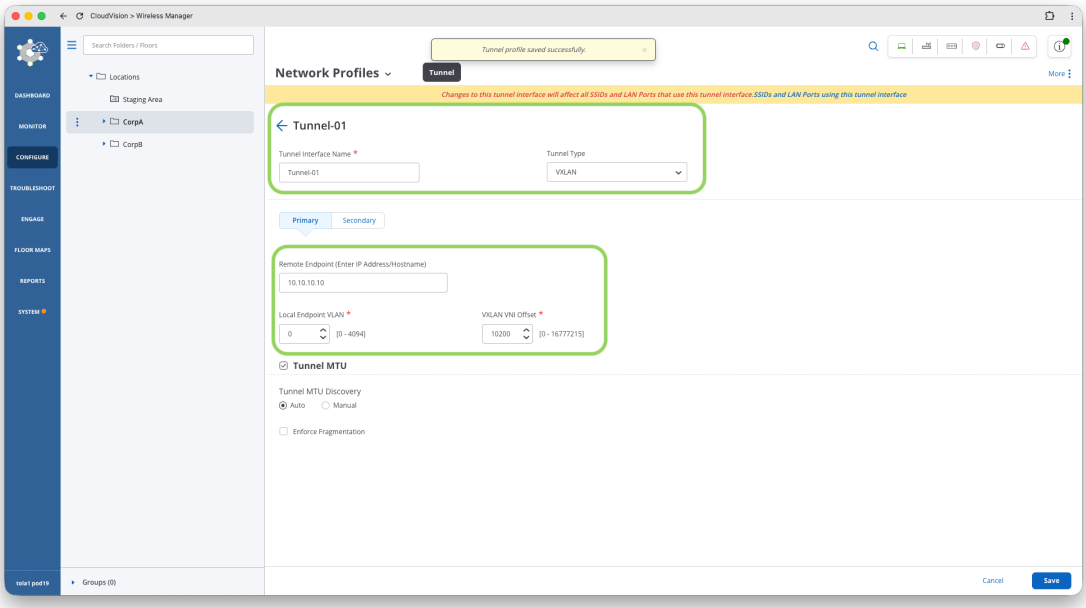
## Create Tunnel Profile (CV-CUE)

1. Under the `Configure` section, select `Network Profiles > Tunnels`
2. Select the correct location in your hierarchy
3. Select `Add Tunnel Interface`

**Network Profiles**

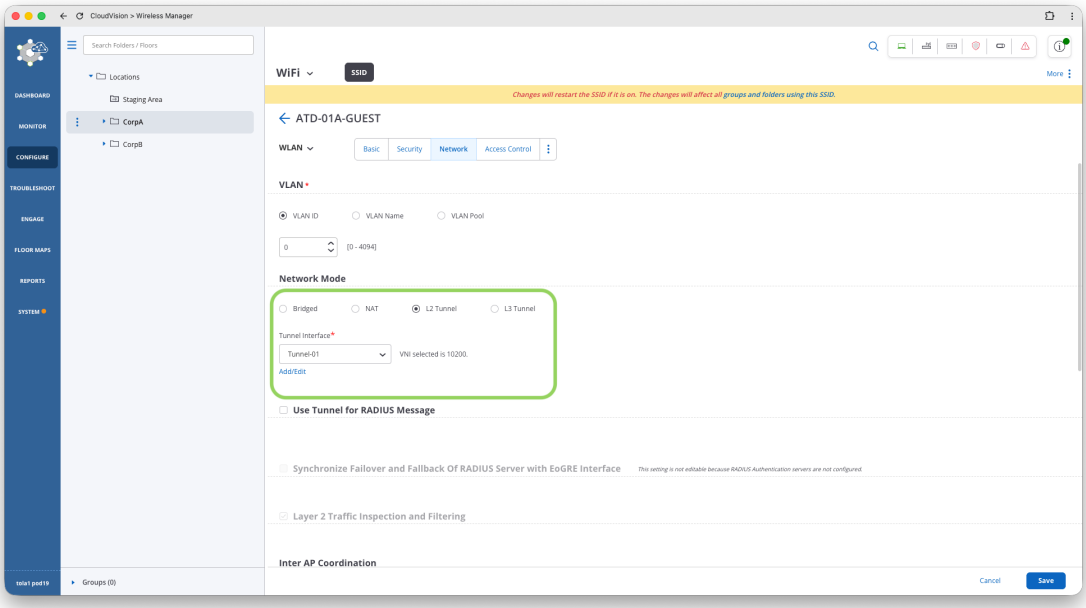
| Setting | Value |
| --- | --- |
| Tunnel Interface Name | tunnel-## |
| Tunnel Type | `VXLAN` |
| Remote Endpoint | 10.10.10.10 |
| Local Endpoint VLAN | 0 |
| VXLAN VNI Offset | 10200 |

4. Select `Save`




## Assign SSID to Tunnel Profile

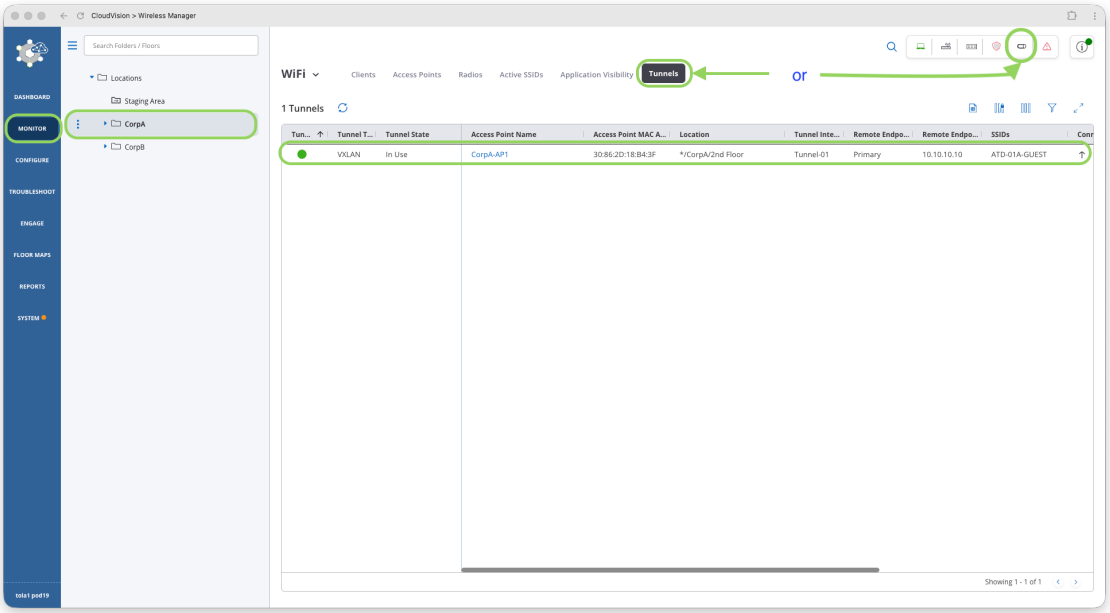
1. Under the `Configure` section, select `Wifi`
2. Select the correct location in your hierarchy
3. Select the pencil to edit Guest SSID
4. Select `Network`
5. Change `Network Mode` to `L2 Tunnel`
6. Select the tunnel-##
7. Select `Save`




# Monitor Tunnels

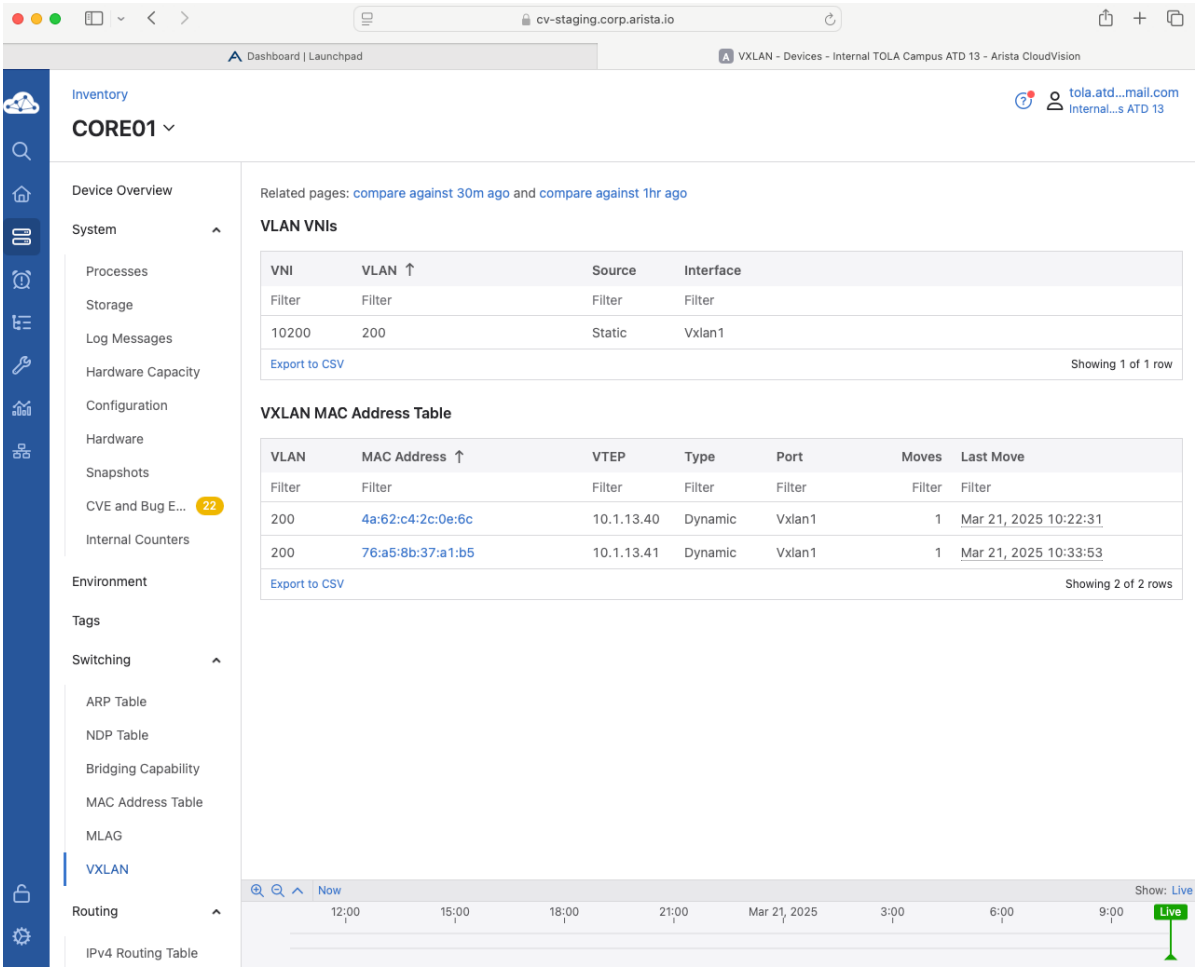
1. Under the `Monitor` section, select `Wifi`
2. Verify your location in the hierarchy
3. Select `Tunnels` and verify it is up

**Notice the client's Guest IP Address is now using network 10.1.200.##**




## Review VXLAN on CORE01 (Instructor CVAAS)

Screenshot shows VLAN200 mapped to VNI10200 and VXLAN clients with VLAN200 IPs




## Additional References

- Captive Portal Configuration
- Guest AP Tunnel
- Configuring RADIUS
- Configuring RadSec

## 🎉 CONGRATS! You have completed this lab! 🎉

**LET'S GO TO THE NEXT LAB!**

2026-06-10